DETECTION OF FAKE AND AI-GENERATED PRODUCT REVIEWS USING DEEP LEARNING: A COMPARATIVE STUDY OF FOUR ARCHITECTURE FAMILIES

Interim report on
Deep Learning Project
[ICT-4442]

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PART B : INTERIM REPORT (Phase 2)

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Title of the Project (confirmed / updated from Synopsis): Detection of Fake and AI-Generated Product Reviews Using Deep Learning: A Comparative Study of Four Architecture Families (unchanged from the synopsis)

GitHub Repository Link: https://github.com/RajRudra06/DL_Project

1. Literature Review

Paper (Author, Year)	Method	Dataset	Key Result	Relevance to Project
Jindal & Liu, 2008 [1]	Logistic regression on review, reviewer and product features, with duplicate reviews used as spam examples	5.8M Amazon reviews, 2.14M reviewers	AUC 98.7% for type 2 and 3 spam, 78% for type 1. Text-only features gave 63%	First large study of opinion spam. Text alone was a weak signal
Ott et al., 2011 [2]	Linear SVM on LIWC and n-gram features	400 truthful and 400 deceptive hotel reviews, the deceptive ones written by crowd workers	89.8% accuracy. Human judges scored 53 to 62%	Source of the hotel corpus reserved for the cross-domain test. Classical baseline behind TF-IDF + MLP
Kim, 2014 [3]	CNN with filters of different widths over pre-trained word vectors	7 sentence-classification benchmarks	Best published result on 4 of 7 tasks with little tuning	Design followed by Text-CNN (filter widths 3, 4, 5)
Ren & Zhang, 2016 [4]	CNN sentence vectors combined by a gated RNN with attention	Hotel, restaurant and doctor reviews	Beat feature-based methods in all three domains	Neural features beat hand-built ones, which supports BiLSTM + Attention
Sanh et al., 2019 [5]	BERT distilled during pre-training with language-modelling, distillation and cosine losses	BERT's pre-training corpus, evaluated on GLUE	40% smaller and 60% faster, keeps 97% of BERT's language understanding	The encoder fine-tuned in this project. Its small size is why cost is part of the comparison
Salminen et al., 2022 [6]	GPT-2 writes the fakes and a fine-tuned RoBERTa detects them	20,000 GPT-2 and 20,000 original Amazon reviews	97% precision and recall (F1 0.97). Human raters 55.36% accuracy	The primary dataset and the reference result to approach with a smaller model
Mitchell et al., 2023 [7]	DetectGPT: zero-shot test based on the curvature of the generator's log-probability	Machine-generated news text (GPT-NeoX in the headline result)	AUROC 0.95 against 0.81 for the best earlier zero-shot method	Needs the generator's probabilities, which a review platform lacks, so this project trains supervised models
AlQadi et al., 2025 [8]	Fine-tuned DistilBERT with LIME and SHAP explanations	2,000 Yelp restaurant reviews, 1,000 human and 1,000 rephrased by ChatGPT	97.25% test accuracy, F1 97.56%	Closest earlier work to the DistilBERT model here. Their set has 2,000 reviews and ours about 40,000
Geetha et al., 2025 [9]	DeBERTa tuned with Monarch Butterfly Optimization	Amazon fake reviews (21k), Fake Reviews (40k), Deceptive Opinion Spam (1.6k, augmented to 16k)	98% accuracy, 98% precision, 97% recall, 97% F1	Uses the same 40k dataset, so a direct number to compare with. Uses DeBERTa instead of DistilBERT
Liu et al., 2026 [10]	FraudSquad: language-model text embeddings plus a gated graph transformer over the review graph	LLM-written Amazon spam (Llama3, Qwen2, Qwen-DSR1) and human spam (Yelp, ChineseQA)	89.45% to 99.98% across datasets	LLM-written spam can be detected, but this method needs review-graph data that a text-only filter lacks

Comparison. Early work used hand-built features and linear models [1], [2]. Jindal and Liu found that review text alone gave 63% AUC, and Ott et al. reached 89.8% accuracy on a small corpus of crowd-written fakes. Neural models then learned the features themselves [3], [4]. The target has since moved from human-written to machine-written fakes. Salminen et al. [6] report 97% precision and recall for a fine-tuned RoBERTa while human raters score 55.36%, and two 2025 studies report 97 to 98% accuracy with DistilBERT [8] and DeBERTa [9]. Each of these studies fine-tunes one Transformer on its own data, so their figures cannot be lined up against each other, and none measures what a smaller model gives up. Zero-shot [7] and graph-based [10] detectors need the generator's probabilities or links between reviewers, which a text-only filter does not have. This project compares four architecture families under one split and one set of metrics and reports parameter count and latency beside accuracy.

2. Dataset Acquisition & Preprocessing

Dataset. The project uses the Fake Reviews Dataset of Salminen et al. [6], from the Kaggle copy of its OSF release. It contains 40,526 product reviews, each with a category, a star rating, the text and a label. Label 1 marks a computer-generated review written by GPT-2 and label 0 an original Amazon review, and the classes are balanced at 20,294 and 20,232. An exploratory analysis script, eda.py, summarises the cleaned data. Computer-generated reviews average 61.2 words against 73.6 for original ones, while the mean star rating is almost the same for both classes, 4.26 against 4.25. The hotel corpus of Ott et al. [2] is reserved as a cross-domain test set and is not used for training.

Cleaning. The shared loader, data.py, drops rows with a missing text or label, strips surrounding whitespace and removes exact duplicate texts, so that no duplicate can sit in both the training and the test split. This removed 35 rows and left 40,491 reviews, 20,260 computer-generated and 20,231 original. No augmentation is applied.

Split. All models share one stratified 70/15/15 split with random seed 42, which gives 28,343 training, 6,074 validation and 6,074 test reviews. The validation and test splits each hold 3,039 computer-generated and 3,035 original reviews. Vocabularies and features are fitted on the training split, model selection uses the validation split, and the test split is read once for the reported numbers.

Encoding. Each model encodes text in its own way on top of the shared split. TF-IDF + MLP uses 15,000 word unigram and bigram features and 25,000 character n-gram features of length three to five. Text-CNN lower-cases the text, drops punctuation and maps each review to 128 indices in a vocabulary of 20,000. BiLSTM + Attention keeps punctuation marks as tokens and uses a vocabulary of 30,000 and 256 positions. DistilBERT uses its own WordPiece tokenizer and cuts reviews to 128 tokens.

3. Models Implemented So Far (minimum 2 of the proposed 3–4)

Model	Owner (Member)	Status	Test metrics	Notes
TF-IDF + MLP	Vedant Agarwal	Complete	Accuracy 0.9503, macro-F1 0.9503, ROC-AUC 0.9904	5.13M parameters. 240 s on a CPU. 8 epochs
Text-CNN	Ishan Satyanand Thakur	Complete	Accuracy 0.9294, macro-F1 0.9293, ROC-AUC 0.9811	2.12M parameters. 136 s on a GPU. 5 epochs
BiLSTM + Attention	Vedant Totla	Complete	Accuracy 0.9555, macro-F1 0.9555, ROC-AUC 0.9920	4.10M parameters. 1,126 s on a CPU. 10 epochs with early stopping
DistilBERT	Rudra Rajpurohit	Complete	Accuracy 0.9709, macro-F1 0.9708, ROC-AUC 0.9980	66.96M parameters. 930 s on a Colab T4 GPU. 3 epochs

Results. All four models are implemented, trained and evaluated on the 6,074 test reviews through the shared harness, evaluate.py, and compare.py builds the comparison table from their result files. DistilBERT is the most accurate at 97.09%, followed by BiLSTM + Attention at 95.55%, TF-IDF + MLP at 95.03% and Text-CNN at 92.94%. DistilBERT makes 177 errors against 302 for TF-IDF + MLP, but it has thirteen times as many parameters. Every model is least accurate on reviews of 20 words or fewer. A first error analysis also shows that 64.5% of computer-generated test reviews end without closing punctuation, against 9.2% of original ones, and all four models do better on computer-generated reviews that end this way. Each model was trained once, so small gaps may not hold for another seed.

4. Individual Contribution Log (to date)

Member Name	Reg. No.	Task(s) Completed	Signature
Rudra Rajpurohit	230911570	Set up the GitHub repository. Built the shared evaluation harness (evaluate.py), the data loader and split (data.py) and the comparison script (compare.py). Fine-tuned DistilBERT on a Colab T4 GPU and analysed its errors. Researched and compiled the literature review. Wrote the nextSteps.md hand-off file for the team.	Rudra
Vedant Totla	230911526	Built BiLSTM + Attention (bilstm_model.py), with a vocabulary fitted on the training split, validation-based early stopping, test evaluation and export of attention weights for the error analysis.	Vedant
Ishan Satyanand Thakur	230953356	Built Text-CNN (textcnn_model.py) and the tokeniser and embedding utilities (preprocess.py). Added the model results to the repository README and updated nextSteps.md.	Ishan
Vedant Agarwal	230953312	Built TF-IDF + MLP (tfidf_mlp_model.py). Wrote the exploratory data analysis script (eda.py), which produces the dataset summary tables and Fig. 1.	Vedant

5. Risk / Plan for Remaining Work

Remaining work. All four models are complete and the comparison table is in the repository. The cross-domain test on the Ott et al. corpus [2], the third objective of the synopsis, has not been run. It will not be completed before the final submission, so the final report lists it as future work. The final report and the in-class presentation remain.

Risks. Three risks remain. Every model was trained once, so the 0.52-point gap between BiLSTM + Attention and TF-IDF + MLP may not hold for another seed. Training times were measured on a CPU for two models and on a GPU for the other two, so they are compared only within one device type. The ending cue means that part of the measured accuracy reflects how the GPT-2 reviews were generated, and the final report treats this as a limitation.

Use of AI tools. Claude (Anthropic) was used to write scaffolding code for the shared evaluation pipeline and the DistilBERT training script, and to help draft parts of this report. All experiments were run by the team.

References

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